

UNIT V

Memory & Programmable Logic Devices: Digital Logic Families: DTL, DCTL, TTL, ECL & CMOS etc., Fan Out, Fan in, Noise Margin; RAM, ROM, PLA, PAL; Circuits of Logic Families, Interfacing of Digital Logic Families, Circuit Implementation using ROM, PLA and PAL

PART **1** Digital Logic Families

A logic family is a group of digital circuits made using same technology.

Different families have different:

- Speed
- Power consumption
- Noise tolerance
- Cost

1 DTL (Diode Transistor Logic)

Structure:

- Diodes for logic operation
- Transistor for amplification

Working:

- Diodes perform AND/OR logic
- Transistor acts as switch

Features:

- Slower
- Higher power than CMOS
- Older technology

2 DCTL (Direct Coupled Transistor Logic)

Structure:

- Only transistors
- No resistors

Features:

- Very fast
- Poor noise margin
- Not commonly used now

3 TTL (Transistor Transistor Logic)

Most popular older logic family.

Uses:

- Bipolar Junction Transistors (BJT)

Features:

- Moderate speed
- Moderate power
- Good reliability
- 5V operation

Used in:

- 7400 IC series

4 ECL (Emitter Coupled Logic)

Fastest logic family (among BJT).

Features:

- Very high speed
- High power consumption
- Low noise margin

Used in:

- High-speed computing
- Supercomputers (old systems)

5 CMOS (Complementary MOS)

Most important modern logic family.

Uses:

- MOSFET (PMOS + NMOS)

Features:

- Very low power
- High noise margin

- High density
- Used in modern CPUs

Most modern processors use CMOS technology.

TTL vs CMOS (Very Important)

Feature	TTL	CMOS
Technology	BJT	MOSFET
Power	High	Very low
Speed	Moderate	High
Noise Margin	Low	High
Used In	Older systems	Modern CPUs

Fan-In

Fan-in = Number of inputs a gate can handle.

Example:

4-input AND gate → fan-in = 4

Higher fan-in → slower operation.

Fan-Out

Fan-out = Number of gates that one output can drive.

Example:

If output drives 10 gates → fan-out = 10

Too high fan-out → signal weakens.

Noise Margin

Noise margin = Ability to resist unwanted noise.

Formula:

Noise Margin = Input threshold – Output level

Higher noise margin → more reliable circuit.

CMOS has high noise margin.

PART Memory Devices

1 RAM (Random Access Memory)

Characteristics:

- Volatile (data lost when power off)
- Read & write memory
- Fast

Types:

1. SRAM (Static RAM)
 - a. Uses flip-flops
 - b. Very fast
 - c. Expensive
2. DRAM (Dynamic RAM)
 - a. Uses capacitors
 - b. Needs refresh
 - c. Cheap

Used in:

- Main memory of computer

2 ROM (Read Only Memory)

Characteristics:

- Non-volatile
- Data stored permanently

Types:

1. PROM – Programmable once
2. EPROM – UV erasable
3. EEPROM – Electrically erasable

Used for:

- Firmware
- BIOS

PART 3 Programmable Logic Devices (PLDs)

Used to implement logic circuits without building from gates manually.

1 PLA (Programmable Logic Array)

Has:

- Programmable AND plane
- Programmable OR plane

Very flexible.

Can implement any SOP expression.

2 PAL (Programmable Array Logic)

Has:

- Programmable AND plane
- Fixed OR plane

Less flexible but faster.

3 ROM as Logic Device

ROM can implement combinational logic.

Idea:

- Inputs = Address
- Output = Stored logic value

Example:

If function $F(A,B,C)$ needed,

Store truth table in ROM.

PLA vs PAL

Feature	PLA	PAL
AND Array	Programmable	Programmable
OR Array	Programmable	Fixed
Flexibility	High	Medium
Speed	Slower	Faster
Cost	Higher	Lower

PART 4 Interfacing of Digital Logic Families

When connecting two logic families:

Important factors:

- Voltage levels
- Current capacity
- Noise margin

Example:

TTL output → CMOS input

Need to ensure voltage compatibility.

Sometimes:

Pull-up resistor required.

PART 5 Circuit Implementation Using ROM, PLA, PAL

Using ROM

Steps:

1. Write truth table
2. Store output values in memory
3. Address lines = inputs

ROM directly gives output.

Using PLA

Steps:

1. Write SOP expression
2. Program AND plane
3. Program OR plane

Using PAL

Steps:

1. Write SOP form
2. Program AND plane only
3. OR plane fixed

1 TTL vs CMOS ★ Very Important

Feature	TTL	CMOS
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Full Form	Transistor-Transistor Logic	Complementary MOS
Technology	BJT (Bipolar Transistor)	MOSFET (PMOS + NMOS)
Power Consumption	High	Very Low
Speed	Moderate	High
Noise Margin	Low	High
Operating Voltage	Usually 5V	Wide range (3V–15V)
Heat Generation	More	Very less
Used In	Old 7400 ICs	Modern CPUs, microcontrollers

Exam line:

CMOS is preferred because of **low power and high noise margin**.

2 RAM vs ROM ★

Feature	RAM	ROM
Full Form	Random Access Memory	Read Only Memory
Volatile	Yes (data lost when power off)	No (data permanent)
Read/Write	Both read & write	Mostly read only
Speed	Fast	Slower than RAM
Used For	Temporary storage	Firmware storage
Example	Main memory	BIOS

Important line:

RAM is working memory.

ROM is permanent memory.

3 SRAM vs DRAM

Feature	SRAM	DRAM
Storage Element	Flip-flops	Capacitors
Speed	Very fast	Slower
Refresh Required	No	Yes
Cost	Expensive	Cheap
Used In	Cache memory	Main memory

Cache memory uses SRAM because it is fast.

4 PLA vs PAL ★

Feature	PLA	PAL
AND Array	Programmable	Programmable
OR Array	Programmable	Fixed
Flexibility	High	Less
Speed	Slower	Faster
Cost	Higher	Lower

Exam tip:

PLA = More flexible

PAL = Faster and cheaper

5 ROM vs PLA

Feature	ROM	PLA
AND Plane	Fixed	Programmable
OR Plane	Fixed	Programmable
Flexibility	Less	More
Memory Required	Larger	Smaller
Used For	Truth table storage	SOP implementation

6 Fan-In vs Fan-Out

Feature	Fan-In	Fan-Out
Meaning	Number of inputs to a gate	Number of gates driven by one output
Affects	Gate delay	Signal strength
High Value Effect	Slower gate	Output becomes weak
Important In	Gate design	Circuit connection

7 DTL vs TTL

Feature	DTL	TTL
Uses	Diodes + Transistor	Only Transistors
Speed	Slow	Faster
Power	Moderate	Moderate
Popularity	Old technology	Widely used (7400 series)

8 ECL vs CMOS

Feature	ECL	CMOS
Speed	Very high	High
Power	Very high	Very low
Noise Margin	Low	High
Cost	High	Lower
Used In	High-speed systems	Modern processors

9 PAL vs PROM

Feature	PAL	PROM
Purpose	Logic implementation	Memory storage
Reprogrammable	Limited	Once programmable
Use	Digital logic design	Permanent data storage

10 Noise Margin vs Noise Immunity

Feature	Noise Margin	Noise Immunity
Meaning	Voltage tolerance range	Ability to resist noise
Higher Value	More reliable	More stable
High In	CMOS	Low in TTL

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